

1.2.2 LUB property of the real numbers

Let $A \subset \mathbb{R}$ be nonempty and bounded above. Then there exists $\alpha \in \mathbb{R}$ such that

$$\alpha = \sup A \text{ (or lub } A).$$

$\mathbb{Q}$ does not satisfy the Least Upper Bound Property.

Exercise 6. Construct a nonempty subset of $\mathbb{Q}$ that is bounded above but whose least upper bound does not belong to $\mathbb{Q}$.

Answer: Take $A = \{x \in \mathbb{Q} : x^2 < 2\}$, then $\text{lub } A = \sqrt{2} \notin \mathbb{Q}$.

This example illustrates that the rational numbers have **gaps**, whereas the real numbers do not.

The above result is known as the **Completeness Property** (or **Least Upper Bound Property**) of the real numbers. It is one of the fundamental properties that distinguishes the real number system from the rational number system. In this course, we shall accept this property without proof and use it throughout our study of calculus. A proof of this result requires a rigorous construction of the real numbers, for example, using Dedekind cuts or equivalence classes of Cauchy sequences of rational numbers, which we shall not discuss in this course.

Theorem 2. The set $\mathbb{N}$ is not bounded above in $\mathbb{R}$.

Proof. Suppose, for contradiction, that $\mathbb{N}$ is bounded above in $\mathbb{R}$. Then, by the Completeness Property, the set $\mathbb{N}$ has a least upper bound, say α .

Since $\alpha - 1$ is not an upper bound of $\mathbb{N}$, there exists $n \in \mathbb{N}$ such that

$$n > \alpha - 1.$$

Hence,

$$n + 1 > \alpha.$$

As $n + 1 \in \mathbb{N}$, this contradicts the fact that α is an upper bound of $\mathbb{N}$.

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. □

Theorem 3 (Archimedean Property). For every $x, y \in \mathbb{R}$ with $x > 0$, there exists

$n \in \mathbb{N}$ such that

$$nx > y.$$

Proof. Proof by contradiction. Suppose that

$$nx \leq y, \quad \forall n \in \mathbb{N}.$$

Then

$$n \leq \frac{y}{x}, \quad \forall n \in \mathbb{N},$$

since $x > 0$. Thus $\frac{y}{x}$ is an upper bound of $\mathbb{N}$, contradicting the previous theorem. Hence there exists $n \in \mathbb{N}$ such that

$$nx > y.$$

□

Exercise 7. Prove that the set of integers $\mathbb{Z}$ is neither bounded above nor bounded below in $\mathbb{R}$.

Recall. For every $x \in \mathbb{R}$, there exists a unique integer $\lfloor x \rfloor$ such that

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

The integer $\lfloor x \rfloor$ is called the *floor* of x . Proving this result using the Archimedean Property is a fun exercise.

Theorem 4 (Density of $\mathbb{Q}$ in $\mathbb{R}$). Let $a, b \in \mathbb{R}$ be such that $a < b$. Then there exists a rational number $r \in \mathbb{Q}$ satisfying

$$a < r < b.$$

Proof. Since $b - a > 0$, the Archimedean Property guarantees the existence of $n \in \mathbb{N}$ such that

$$n(b - a) > 1.$$

Let

$$k = \lfloor na \rfloor, \quad m = k + 1.$$

Then

$$k \leq na < k + 1 = m,$$

and hence

$$na < m.$$

We claim that $m < nb$. Suppose, on the contrary, that $m \geq nb$. Then

$$1 = (k + 1) - k = m - k \geq nb - na = n(b - a) > 1,$$

which is impossible. Therefore,

$$m < nb.$$

Consequently,

$$na < m < nb.$$

Dividing throughout by the positive integer n , we obtain

$$a < \frac{m}{n} < b.$$

Since $m, n \in \mathbb{Z}$ and $n > 0$, the number

$$r = \frac{m}{n}$$

is rational. This completes the proof. $\square$

Corollary 1. *Let $a, b \in \mathbb{R}$ be such that $a < b$. Then there exists an irrational number $t \in \mathbb{R} \setminus \mathbb{Q}$ satisfying*

$$a < t < b.$$

Proof. Since

$$a - \sqrt{2} < b - \sqrt{2},$$

the density of $\mathbb{Q}$ in $\mathbb{R}$ guarantees the existence of a rational number r such that

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

Adding $\sqrt{2}$ throughout, we obtain

$$a < r + \sqrt{2} < b.$$

We claim that $r + \sqrt{2}$ is irrational. Indeed, if $r + \sqrt{2}$ were rational, then

$$\sqrt{2} = (r + \sqrt{2}) - r$$

would be the difference of two rational numbers, and hence rational, a contradiction since $\sqrt{2}$ is irrational. Therefore,

$$t = r + \sqrt{2}$$

is an irrational number satisfying

$$a < t < b.$$

This completes the proof. $\square$

Absolute Value. For $x \in \mathbb{R}$, the *absolute value* of x is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

We now recall some useful properties of the absolute value. For all $a, b \in \mathbb{R}$,

1. $|a| \geq 0$ and $|a| = 0$ iff $a = 0$. Moreover, $-|a| \leq a \leq |a|$.
2. Let $\varepsilon > 0$. Then $|a - b| < \varepsilon$ iff $a \in (b - \varepsilon, b + \varepsilon)$.
3. $|ab| = |a||b|$ and $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, provided $b \neq 0$.
4. Triangle Inequality: $|a + b| \leq |a| + |b|$.
5. Reverse Triangle Inequality: $||a| - |b|| \leq |a - b|$.
6. $\max\{a, b\} = \frac{a+b+|a-b|}{2}$ and $\min\{a, b\} = \frac{a+b-|a-b|}{2}$.